

System Overview, Conventions & Hyperparameter Atlas

What the Vol-Fitter does, the pipeline, and the master index of every knob
Vol-Fitter Technical Notes, No. 00

Vol-Fitter

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Abstract

This is the map of the territory. The Vol-Fitter is an implied-volatility surface fitter in the spirit of a professional vola-fitter, with one differentiating capability: it *extrapolates* sparse smile observations to a full universe of smiles, across expiries *and* assets, by propagating signal through a graph whose nodes are smiles (underlying, T). This note states the market and normalization conventions shared by the whole series, walks the end-to-end compute pipeline (fetch $\rightarrow$ de-Americanize $\rightarrow$ forwards $\rightarrow$ fit $\rightarrow$ views $\rightarrow$ graph), catalogues what the application can do, and ends with a *master hyperparameter atlas* cross-linked to the topic notes that derive each knob. It is the front door: read it once, then any other note in the series stands alone.

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1 What the application is

At its core the Vol-Fitter solves one problem repeatedly — *given noisy, sparse option quotes for one expiry of one underlying, produce a smooth, arbitrage-free implied-volatility smile* — and then layers three ambitions on top of it:

1. **A choice of models, all arbitrage-aware.** Four smile representations share one calibration spine: the *LQD* log-quantile-density model (Note 01), *SVI* / *SVI-JW* (Note 02), the *Multi-Core Sigmoid (MCS)* family (Note 03), and a full *piecewise-affine local-volatility* surface (Note 04). The first three are slice parametrizations that are butterfly-free by construction or by penalty; the fourth is a genuine local-volatility grid solving the Dupire PDE.
2. **A professional desk workflow.** Real bid/ask handling (fit to mid, to the bid–ask band, or to a haircut band), var-swap targets, density-corrected weighting, event time dilation, dividend and forward treatment, spot–vol dynamics, a Bayesian prior-persistence machine that decides, quote by quote, whether to trust the market or the saved prior, and a temporal observation filter (a per-handle Kalman/MAP estimator, Note 15) that denoises noisy or contradictory quotes without ever confusing noise with data gaps.
3. **Graph smile-extrapolation.** The headline differentiator (Note 14): a transported prior plus the lit-node calibration innovation are propagated through an optimal-transport-regularized Bayesian graph to reconstruct the *dark* nodes nobody quoted.

Heuristic

The unifying idea is that a volatility surface is not a curve to be drawn but a *family of probability laws* to be inferred — one risk-neutral law per (underlying, T) node — subject to no-arbitrage couplings (butterfly within a slice, calendar across expiries) and to a Bayesian prior that ties the whole universe together. Every design choice in the app follows from taking that sentence literally.

Invariant

The system-wide contracts every note protects, stated once:

1. **Arbitrage-cleanliness is structural where possible, measured where not.** LQD and the LV surface are clean by construction; SVI/MCS are fenced by penalties that are exactly zero on admissible slices; and the residual scheme noise is reported by explicit diagnostics, never assumed away.
2. **Every feature is strictly additive.** Bands, var-swap targets, calendar coupling, the event clock, priors, betas, the wing penalty, the observation filter: each is byte-identical to its absence when off — and that byte-identity is test-locked, not asserted.
3. **Numbers are measured, not copied.** Figures and tables are generated fresh by the production code; reference listings are executed against the modules they distill; claims carry test anchors.
4. **Versioned invalidation is scoped.** Editing one name’s rate, dividends, events or prior bumps that ticker’s version and refits that name — never the universe.

Figure 1 shows the two structural guarantees on a four-expiry synthetic surface fitted by the production LQD calibrator: the implied densities are all non-negative (no butterfly arbitrage) and the ATM total-variance term structure is non-decreasing (no calendar arbitrage).

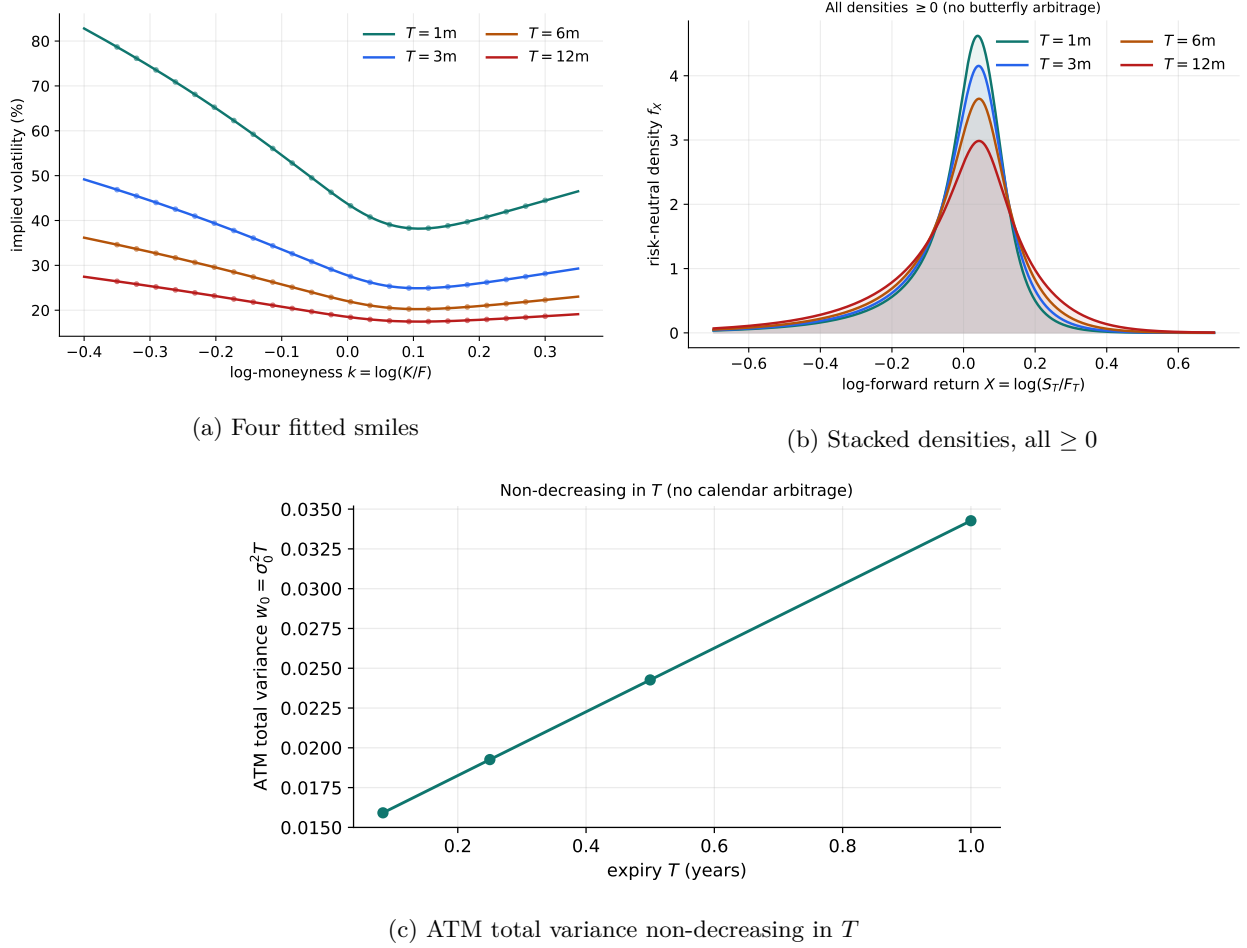


Figure 1: The two no-arbitrage guarantees, produced end to end by the production fitter on a synthetic but calendar-monotone surface.

2 Market conventions and normalization

These conventions hold across the entire series; individual notes assume them without restating.

Definition 1 (Forward measure and normalization). For expiry T with forward price F_T , work under the T -forward measure and in forward-normalized, undiscounted units. The log-moneyness is $k = \log(K/F_T)$, the log-forward return is $X_T = \log(S_T/F_T)$, and normalized call/put prices are $C_T(k) = \mathbb{E}^T[(e^{X_T} - e^k)^+]$ and $P_T(k) = \mathbb{E}^T[(e^k - e^{X_T})^+]$, with the martingale condition $\mathbb{E}^T[e^{X_T}] = 1$.

Definition 2 (Total variance and the Black map). Total implied variance is $w(k, T) = \sigma_{BS}^2(k, T) T$. The normalized Black call is $B(k, w) = \Phi(d_+) - e^k \Phi(d_-)$ with $d_{\pm} = -k/\sqrt{w} \pm \sqrt{w}/2$. Implied total variance is the unique w solving $B(k, w) = C_T(k)$; the fitter inverts it by a vectorized scheme rather than a scalar root-find per strike.

This one primitive underlies every model in the series; in normalized form it is a handful of lines (matches production to 10^{-12}):

Listing 1: The normalized Black call $B(k, w)$ — the shared pricing primitive (distilled from `core/black.py`).

```

1  def black_call(k, w):                                     # k = log(K/F), w = total
    variance sigma^2 T
2      sq = np.sqrt(w)
3      dp = -k/sq + 0.5*sq                                  # d_+; d_- = d_+ - sqrt(w)
4      price = norm_cdf(dp) - np.exp(k)*norm_cdf(dp - sq)
5      return np.where(w > 1e-12, price, np.maximum(1 - np.exp(k), 0.0))
    # intrinsic at w->0

```

Three conventions deserve emphasis because they recur:

- **Two clocks.** Calendar time t (year-fraction to expiry) and *variance* time τ (calendar time dilated by scheduled events) are kept separate. Fits live on τ ; implied volatility is reported as $\sqrt{w/\tau}$, so adding an event *lowers* reported IV while preserving price (Note 11).
- **Forward-normalized everything.** Prices, strikes and densities are quoted relative to F_T , so a single smile object is spot-independent until a dynamics rule (Note 12) transports it under a spot move.
- **Vega-normalized residuals.** Calibration losses are price residuals divided by Black vega, which makes the objective behave like a volatility error while keeping every iterate a valid price (Notes 01, 04, 07).

3 The compute pipeline

Every smile the app shows is the output of the pipeline in fig. 2. The stages are deliberately decoupled and independently cached so that, e.g., editing a fit penalty never re-pulls data or re-runs de-Americanization.

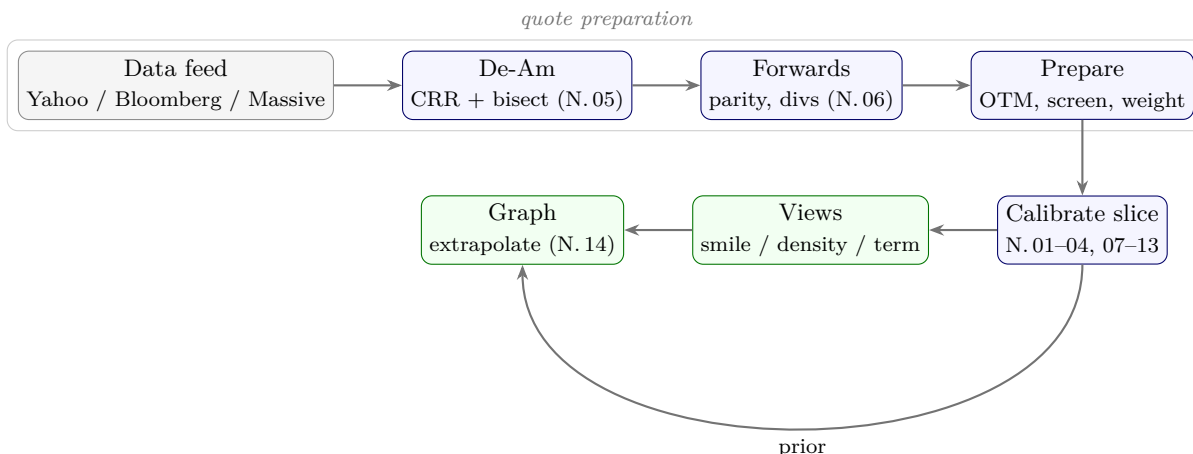


Figure 2: The end-to-end compute spine. Quote preparation (de-Americanize, forwards/dividends, OTM selection, static screen, weighting) feeds the per-slice calibration; the calibrated lit nodes drive both the viewer and the graph extrapolator, which reconstructs the dark nodes.

Stage by stage.

1. **Data feed.** A provider registry exposes Yahoo (US options), Bloomberg (xbbg; US + non-US indices/equities), and Massive (Polygon/OPRA), plus a synthetic source; an as-of selector serves Live / previous-close / EOD / captured-replay. US-options-only providers and the non-US Bloomberg path are handled transparently.
2. **De-Americanization** (Note 05). Listed equity options are American; each quote is converted to its European-equivalent implied volatility by inverting a Cox–Ross–Rubinstein tree (bisection), with a Numba kernel for wide chains and a content-digest cache so fit-tuning never re-runs it. A wing-only, band-constrained convexity repair then removes the butterfly arbitrage the per-strike inversion can leave in the sparse wings, holding the ATM core fixed.
3. **Forwards and dividends** (Note 06). Put–call parity yields the implied forward and discount (clamped against feed noise); discrete dividends and American de-bias are handled here. One provider pathology is handled upstream of the regression: chains a delayed feed *synthesizes* at zero carry (NBBO gated, so the provider emits Black prices from its own IV marks at $F = \text{spot}$, $D = 1$) are flagged at capture and *pinned* to $F = \text{spot}$, $D = 1$ rather than parity-regressed — regressing quotes that already embed the provider’s carry produced garbage forwards.
4. **Prepare.** Out-of-the-money selection, a static no-arbitrage/wing screen, and the choice of observation weights (Note 07).
5. **Calibrate slice.** The chosen model is fitted to the prepared quotes under the active objective: mid / bid–ask band / haircut (Note 07), optional var-swap target (Note 08), calendar coupling (Note 10), event clock (Note 11), and Bayesian prior persistence (Note 13). On commit, the fitted ATM handles can additionally pass through the observation Kalman filter (Note 15; **off/overlay/active**, **off** byte-identical to the feature never existing): a per-handle temporal state estimator that weighs today’s noisy observation against the previous filtered state transported under the Note 12 dynamics.

6. **Views.** Every smile-derived view — smile, stacked densities, stacked total variance, quantile density, term structure, local-vol surface, SSR scenario — follows the chosen model and is downsampled for the UI.
7. **Graph** (Note 14). The transported prior plus the lit-node innovation propagate through the OT-regularized Bayesian graph to reconstruct dark-node smiles with credible bands.

4 The smile graph

The object that makes the app more than a slice fitter is the *smile graph* (fig. 3): nodes are smiles (underlying, T); edges encode that one node informs another. Calendar edges link expiries of one name; cross-asset edges link an index or sector ETF to its single names. A user marks nodes *lit* (quoted, calibrated) or *dark* (to be reconstructed). The graph carries the three ATM handles (σ_0, s_0, κ_0) of each node as a Gaussian signal and propagates the lit innovation to the dark nodes; Note 14 develops the optimal-transport-regularized Bayesian solver behind it, and the leave-one-out backtest that measures its skill. That skill is now measured at scale: on the 25-asset benchmark across three historical regimes ($\sim 47k$ held-out scores), neighbour-supported indexes gain +10 to +76 ATM vol bps over the transported-prior baseline and ETFs +3 to +7; fully-dark single names behind lit indexes/ETFs — the product case — gain +7.9 to +14.2 bps in the August-2024 spike and +3.8 to +7.2 fully out-of-sample in the October-2022 bear, with unbiased standardized residuals. In the calm July-2023 regime the dark-name skill is ≈ 0 (never negative): single-name moves there are earnings-idiosyncratic, and the graph carries shared repricing, not idiosyncratic news. The one flagged weakness is the calm regime’s dark-name bands ($\text{std } \zeta \approx 1.9$, overconfident) — an open follow-up recorded in Note 14.

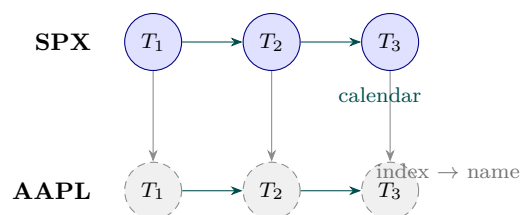


Figure 3: A fragment of the smile graph: lit index nodes (solid) inform dark single-name nodes (dashed) across both calendar and cross-asset edges. The graph posterior reconstructs the dark smiles from the lit innovation.

5 Capability catalogue

Table 1 indexes what the app can do against the note that documents the mathematics. It doubles as a reading order.

Table 1: Capability $\rightarrow$ note index.

Capability	Summary	Note
LQD smile model	Arbitrage-free-by-construction log-quantile density; analytic Jacobian	01

Capability	Summary	Note
SVI / SVI-JW	Raw and jump-wing SVI with no-arb penalty and Lee cap	02
Multi-Core Sigmoid (MCS)	Sigmoid base + zero-wing hats for WW/event smiles	03
Local volatility	Piecewise-affine Dupire surface; GN/Numba calibration	04
De-Americanization	CRR tree + bisection, Numba kernel, cache digest, wing convex repair	05
Forwards / dividends	Parity forward + discount clamp, discrete divs	06
Weighting & bid-ask	Equal / time-value-density weights; band/haircut fit	07
Variance-swap targets	Replication, LQD closed form, source-PDE	08
Wings & Lee bounds	Tail control unified across models	09
Calendar arbitrage	Convex-order soft hinge, model-agnostic	10
Event variance clock	Dual calendar/variance time, day-weighted events	11
Spot-vol dynamics	Sticky strike/moneyness/local-vol, SSR transport	12
Prior persistence	7-mode Bayesian menu, precision gate, two-pass	13
Graph extrapolation	Transported prior + directed increments + back-test	14
Observation Kalman filter	Per-handle temporal denoising of the ATM handles; one-stage MAP active mode	15

Viewer and data capabilities (engineering, not derived here). The workspace is organized into Parametric, LocalVol, Forwards, Options, Graph and Universe tabs: charts of prior/current fit vs. quote bands in normalized or fixed strike, quantile and density functions, term structure and event-dilated calendar in vol and variance, var-swap levels, interactive quote select/erase/amend, universe selection, lit/dark node maps, and the graph editor. Data plumbing (providers, as-of replay, dividend editor, persistence of named universes and fit history) is described in the project README rather than in this mathematical series.

6 The design rules that recur

Reading the series end to end, a handful of design rules appear again and again — each learned from a production incident, each now documented as a *case file* in its note. They are worth stating once as the app’s engineering philosophy.

1. **The confinement principle** (Note 09). A constraint comparing *two* curves is sampled only where data pins both; a constraint intrinsic to *one* curve holds everywhere, wings included. Four instances: the calendar floor (confined, Note 10), the LV convex wing (confined, Note 04), the MCS Durrleman penalty (extended, Note 03), the de-Am convexity repair (extended, but its *authority* confined to the quoted band, Note 05).
2. **Trust the well-identified parameter.** The parity forward keeps the price level and refuses the noisy slope (Note 06); the prior gate trusts the market exactly where quotes identify a

feature and yesterday exactly where they do not (Note 13).

3. **Persist shape, not level.** Operators and factors are level-invariant baskets, so an overnight jump is never damped (Note 13); the graph propagates *increments*, so “no change” is the default dark-node answer (Note 14).
4. **Route each model through its native object.** The var-swap is a closed form for LQD, arithmetic for SVI/MCS, a source PDE for LV (Note 08); the calendar constraint is the asset share for LQD and a total-variance hinge for the overlays (Note 10). Uniform implementations across non-uniform models are a trap.
5. **Measure first, fix second.** The short-dated LV failure was convicted by per-expiry counts before any model change (Note 04); the MCS wing penalty sits where the arb census located the violations (Notes 03, 09); the temporal backtests score priors, the graph and the observation filter against honest held-out baselines (Notes 13–15).

Table 2 indexes the production incidents documented across the series — the fastest way to learn the system is arguably to read these first.

Table 2: Case-file index: production incidents and their lessons.

Incident	Lesson	Note
The six-day weekly that broke the LV strike grid (108 $\rightarrow$ 23.5 bp)	measure first; per-expiry coverage floors	04
The convex wing that flattened SPY (25.7 $\rightarrow$ 2.6 bp)	constraints stay off quoted territory	04
The reverted global de-Am repair (ATM smile gap)	repairs need confined authority	05
The delayed feed that gapped the ATM smile (discount clamp; failed rate-anchor)	trust the well-identified parameter	06
The fit that took minutes (var-swap replication in the Jacobian)	native-object routing	08
The MCS that invented a put wing (R3 $\times$ R6 ablation, 92/25/749/225 bp)	input repair and output penalty compose	09, 03
The phantom calendar (NVDA/SPY flattened; 151 $\rightarrow$ 0 bp reproduced)	the confinement principle	10, 09
The jump the strike anchor damped (operators persist shape)	persist shape, not level	13
The backtest that confirmed the hybrid default ($\sim$ 32 bp on the held-out wing, 1117 nodes)	defaults are chosen by held-out evidence	13
The transient-name topology defect (cross-asset skill exactly 0.000)	exact zeros are disconnection, not damping	14
The close-strike contradiction vs. the true gap (gains 0.80/0.72/0.03)	noise is not a gap: filter and prior split the work	15
The off-diagonal blow-up on coarse chains (3–28 vol points)	per-handle updates; caps do not police cross-gains	15

7 How the notes are organized

Each note is standalone and follows the same skeleton, codified in `Docs/notes/STYLE_GUIDE.md`: an opening that states the *production problem* and the *invariants* being protected (the amber box near the top of every note), a theorem-shaped body whose central object gets the note’s single boxed equation, a worked example or *case file* told as a production incident, a *traceability table* tying each claim to its module and locking test, and three appendices — **A, a hyperparameter atlas** (every knob, surfaced or hidden, with its default), **B, performance** (measured optimizations, including the rejected ones), and **C, a reference implementation** (≤ 50 lines, executed against the production module before being committed). Figures run through the shared `figures/style.py` and their captions state the lesson, not the axes. Notes cross-reference each other (this note’s table 1 is the index) but never depend on one another’s internals. Numbers and plots are *measured*, not copied: where a fresh run differs from an older design note, the fresh run wins.

A Master hyperparameter atlas

The complete index of tunable parameters across the app: surfaced (the user-facing `FitSettings` and `OptionsSettings`) and hidden (internal constants). Each row points to the note that derives its role; defaults are the shipped values. This is a directory, not a derivation — consult the referenced note for meaning.

Table 3: Surfaced parameters.

Parameter	Default	Role	Note
<code>nOrder</code>	6	LQD Legendre order	01
<code>regLambda / regPower</code>	10^{-6} / 1	LQD high-order ridge	01
<code>barrierCenter / barrierScale</code>	0.90 / 50	LQD A_R barrier	01
<code>nCores</code>	2	MCS hat count (schema-clamped at 2)	03
<code>sigmoidRidge</code>	10^{-2}	MCS hat-amplitude ridge	03
<code>sivWingPenaltyPct</code>	100	MCS put-wing Durrleman penalty (0 = off)	03
<code>sviPenaltyWeight</code>	10^3	SVI no-arbitrage penalty	02
<code>leeSlopeMax</code>	2.0	SVI Lee wing-slope cap	02, 09
<code>gridXNodes</code>	12	LV strike-grid vertices	04
<code>gridXMinPerExpiry</code>	8	LV short-dated coverage floor	04
<code>gridTNodes</code>	10	LV maturity-grid vertices	04
<code>gridRegLambda / gridRegRho</code>	10^{-2} / 1	LV roughness penalty / ratio	04
<code>convexWingWeight</code>	10^3	LV wing-convexity penalty (<code>convexWing</code> off by default; tail-confined)	04
<code>frontTieWeight</code>	10^{-2}	LV front-end tie-to-ATM	04
<code>lvVolCapMult</code>	3.0	LV local-vol cap multiple	04
<code>weightScheme</code>	equal	equal / tv-density weights	07
<code>fitMode</code>	mid	mid / bidask / haircut	07
<code>haircut</code>	0.005	band haircut (vol)	07
<code>midAnchorWeight</code>	0.05	band mid anchor	07
<code>varSwapWeightPct</code>	10%	var-swap budget	08
<code>calendarWeight</code>	10^6	calendar soft-constraint penalty	10
<code>ssr</code>	2.0	spot-vol SSR / regime	12

Parameter	Default	Role	Note
priorPersistenceMode	hybrid	prior mode selector	13
priorOperatorSet	ATM, RR25, BF25, VS	operators to persist	13
priorOperatorStrengthPct	50%	operator prior budget	13
priorOperatorRequiredPrecision	1.0	operator activation gate	13
priorOperatorGapExponent	1.0	gate exponent γ	13
priorOperatorBandwidth	0.06	operator leg KDE bandwidth	13
priorFactorSet	ATM, skew, curv, VS	smile factors to persist	13
priorFactorStrengthPct	50%	factor prior budget	13
priorTailAnchorStrengthPct	20%	hybrid tail-anchor budget	13
priorAnchorWeightPct	50%	strike-gap prior budget	13
graphKappaScale κ	1.0	graph temporal precision	14
graphEtaScale η	1.0	graph directed smoothness	14
graphLambdaScale λ	0.0	graph OT regularization	14
graphNu ν	0.1	graph UOT source allowance	14
events	—	scheduled event clock	11
observationFilterMode	off	filter off / overlay / active (MAP)	15
filterCovarianceMode	jacobian	measurement-covariance route	15
filterProcessVolBpSqrtDay	30	ATM clock process noise ($\text{bp}/\sqrt{\text{day}}$)	15
filterProcessSkewSqrtDay	0.02	skew clock process noise	15
filterProcessCurvSqrtDay	0.05	curvature clock process noise	15
filterTransportNoiseScale	0.10	process std per unit transport distance	15
filterResidualInflation	on	contradiction reads as measurement noise	15
filterAdaptiveSigma	3	innovation-gated adaptive Q (0 = off)	15
filterMaxGain	1.0	per-handle own-gain safety cap	15
filterResetHours	96	maximum data gap predicted across	15
filterDataOnlyPrepass	off	strictly persistence-free measurement	15

Table 4: Hidden internal constants (selected).

Constant	Default	Role	Note
Z_MAX / N_POINTS	40 / 8001	LQD quadrature grid	01
OPT_N_POINTS	2001	LQD optimization grid	01
EPS_AR	10^{-6}	LQD $A_R < 1$ buffer	01
DEFAULT_N_K	1201	LV PDE strike nodes	04
DEFAULT_DT_MAX	1/400	LV PDE max time step	04
N_RANNACHER	4	LV Rannacher start steps	04
VAR_FLOOR	10^{-6}	Dupire local-variance floor	04
DEFAULT_STEPS	501	CRR scalar tree depth	05
DEFAULT_BATCH_STEPS	192	CRR batch tree depth	05
BATCH_BISECTIONS	24	de-Am bisection sweeps	05
SIGMA_LO / SIGMA_HI	10^{-4} / 4	implied-vol bracket	05
VS_HALF_WIDTH / VS_POINTS	6 / 801	var-swap replication grid	08
CAL_STRIDE	25	calendar constraint stride	10
DAYS_PER_YEAR	365	calendar→variance clock	11
SPREAD_HALF	0.05	precision spread half-width	13
OBS_FRESHNESS_HALFLIFE	3 d	observation freshness decay	13

Constant	Default	Role	Note
TRANSPORT_SCALE	0.10	prior transport precision decay	13
HANDLE_CONFIDENCE	[1, 1, 0.01]	per-handle graph confidence	14
SOURCE_BASE	tiered	prior provenance precision	14
DIAGONAL_UPDATE	on	per-handle scalar Kalman gains	15
RESID_INFLATION_CAP	25	cap on the contradiction inflation	15

Reading the atlas. A surfaced parameter is a *policy* choice — how strong a penalty, which prior mode, how to weight quotes — safe to tune. A hidden constant is tied to an algorithm’s correctness (the A_R buffer, the PDE floor, the tree depth) and is exposed only in the relevant note’s Appendix A, where the consequences of changing it are spelled out. Together they are the full control surface of the fitter.